

Politics of emotions or propaganda?

Natural Language Processing

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Introduction

Goal

To explore how emotional language is used in political texts to influence perception and manipulate audience response, define emotional profile of speakers.

Approach

- fine-tune a transformer-based model on a sentiment analysis task;
- annotate the dataset;
- perform analysis of emotional profile with sentiment rhetorical function definition.

Data

Text transcriptions of US 2020 presidential debates between Joe Biden and Donald Trump.

Research question and methodology

Politicians use emotional language to influence the audience, and build stronger connections with it. By using specific words and phrases in their speech they can evoke fear, love and anger, thus trying to control the reaction and response of the audience.

In order of being able to draw a conclusion about the usage of emotional language the emotional profile of a speaker is created.

In particular, the analysis focuses on:

- emotion flow throughout a single event;
- emotion profile during a specific debate event;
- comparison of emotional profiles across different debate events;
- emotion distribution by topic.

The speech annotation is done using the fine-tuned DistilBERT classifier.

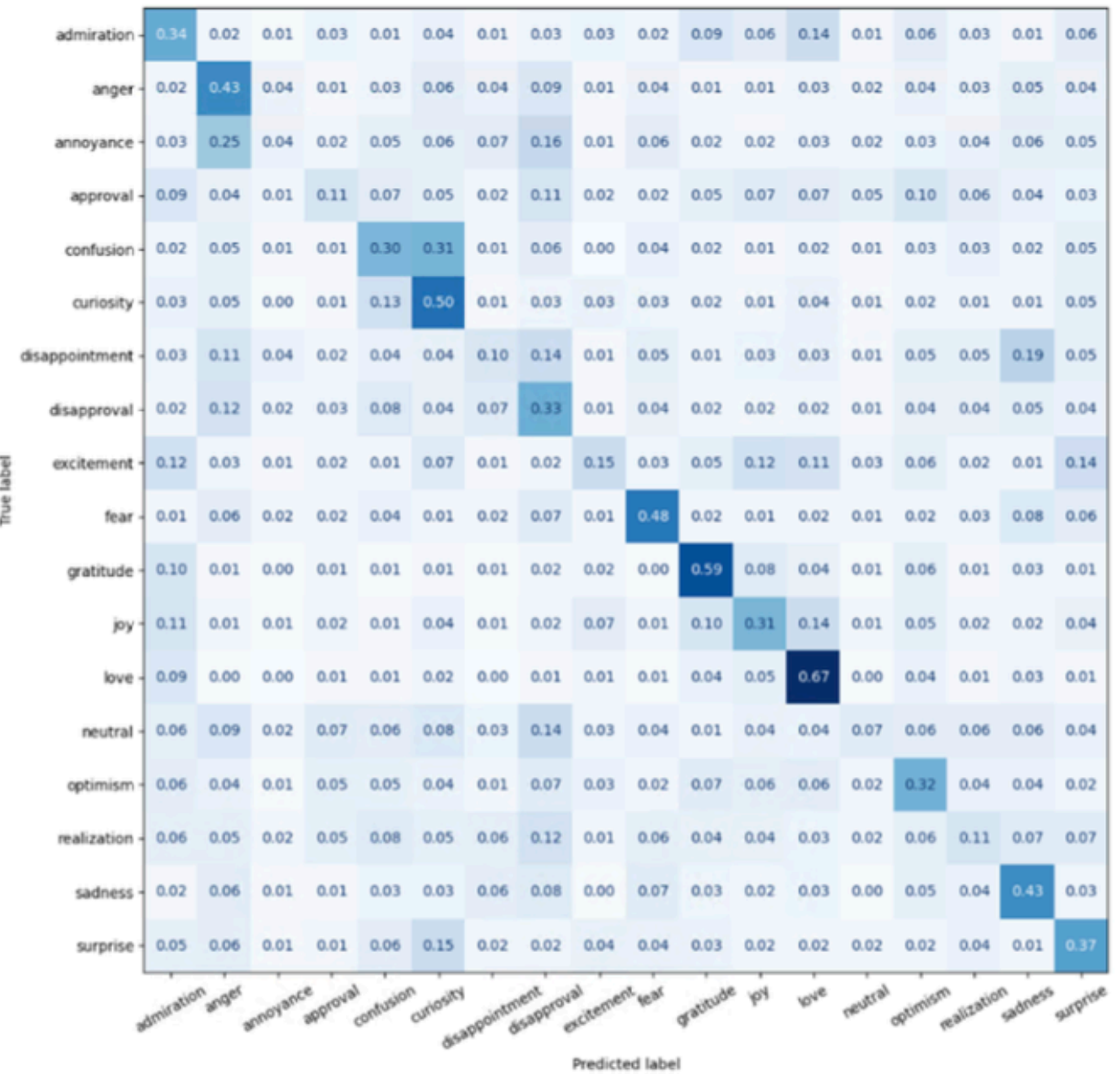
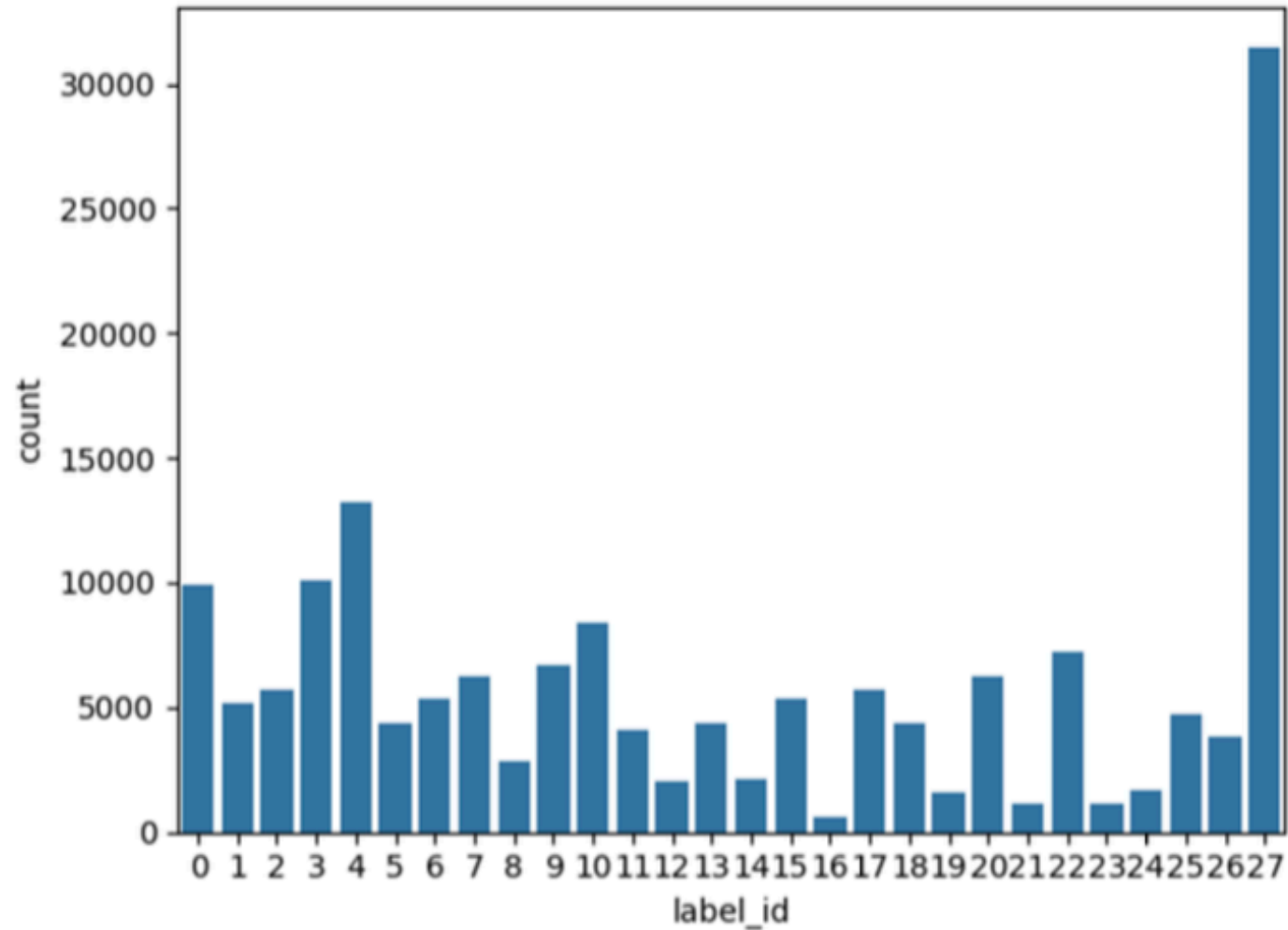
Experimental results

Why DistilBERT?

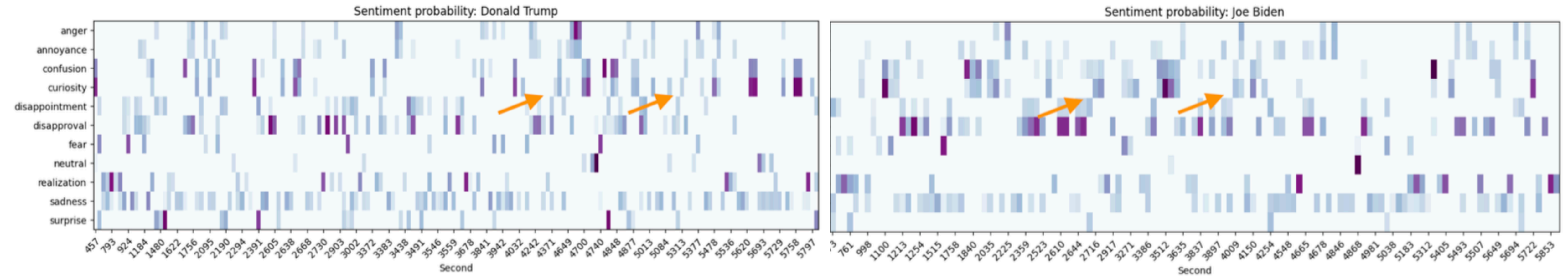
Emotion classification models based on pre-trained BERT architectures have been shown to outperform other approaches while. DistilBERT is pretrained by knowledge distillation to create a smaller model with faster inference and requires less compute to train (and similar performance to a larger model).

Model training

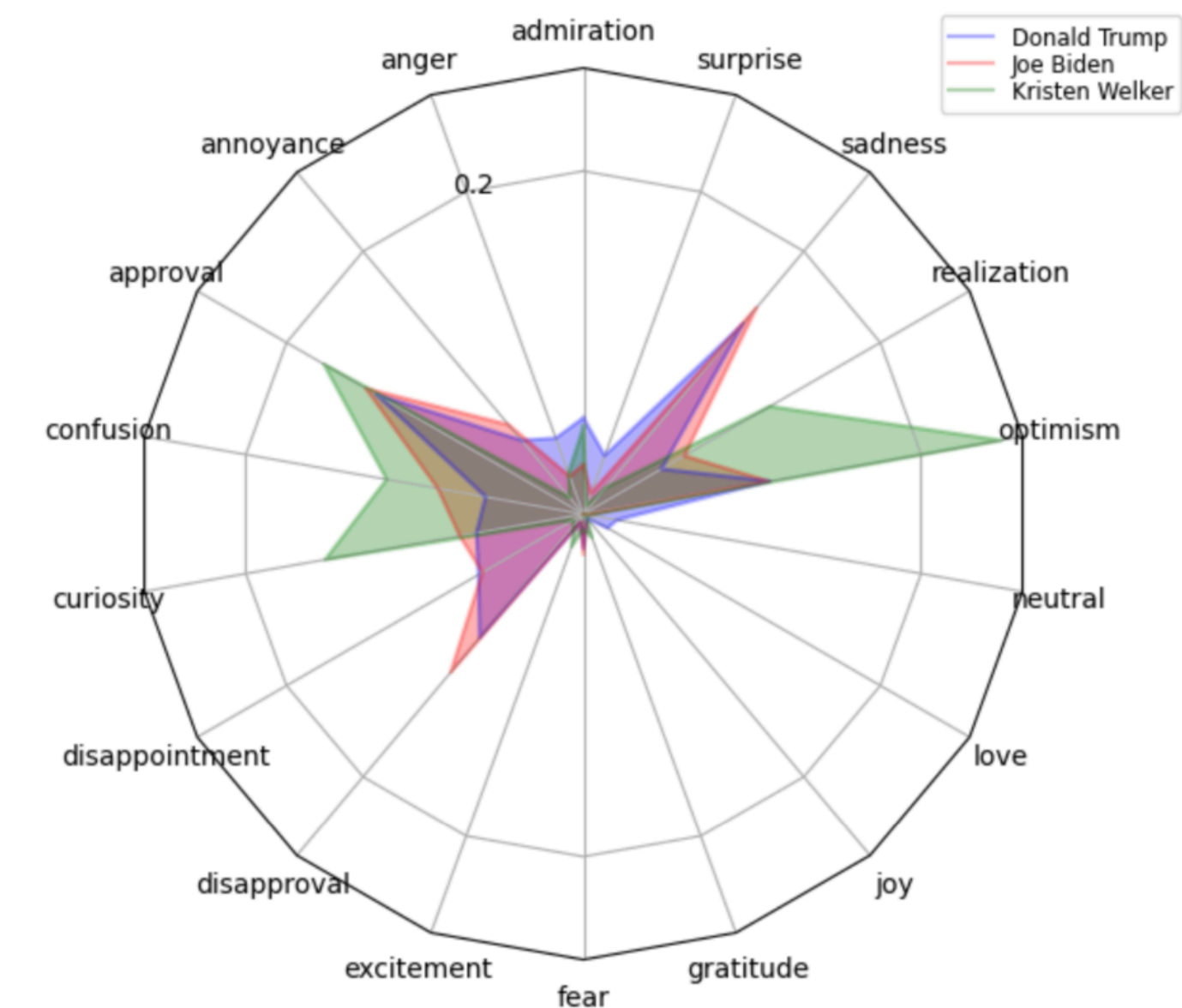
- The GoEmotions dataset: 58 000 Reddit comments, 27 emotion classes
- Optimizer: Adam
- Learning rate: 1e-5
- Epochs: 2
- Accuracy: 0.29



Speakers display mostly similar emotional language with subtle differences

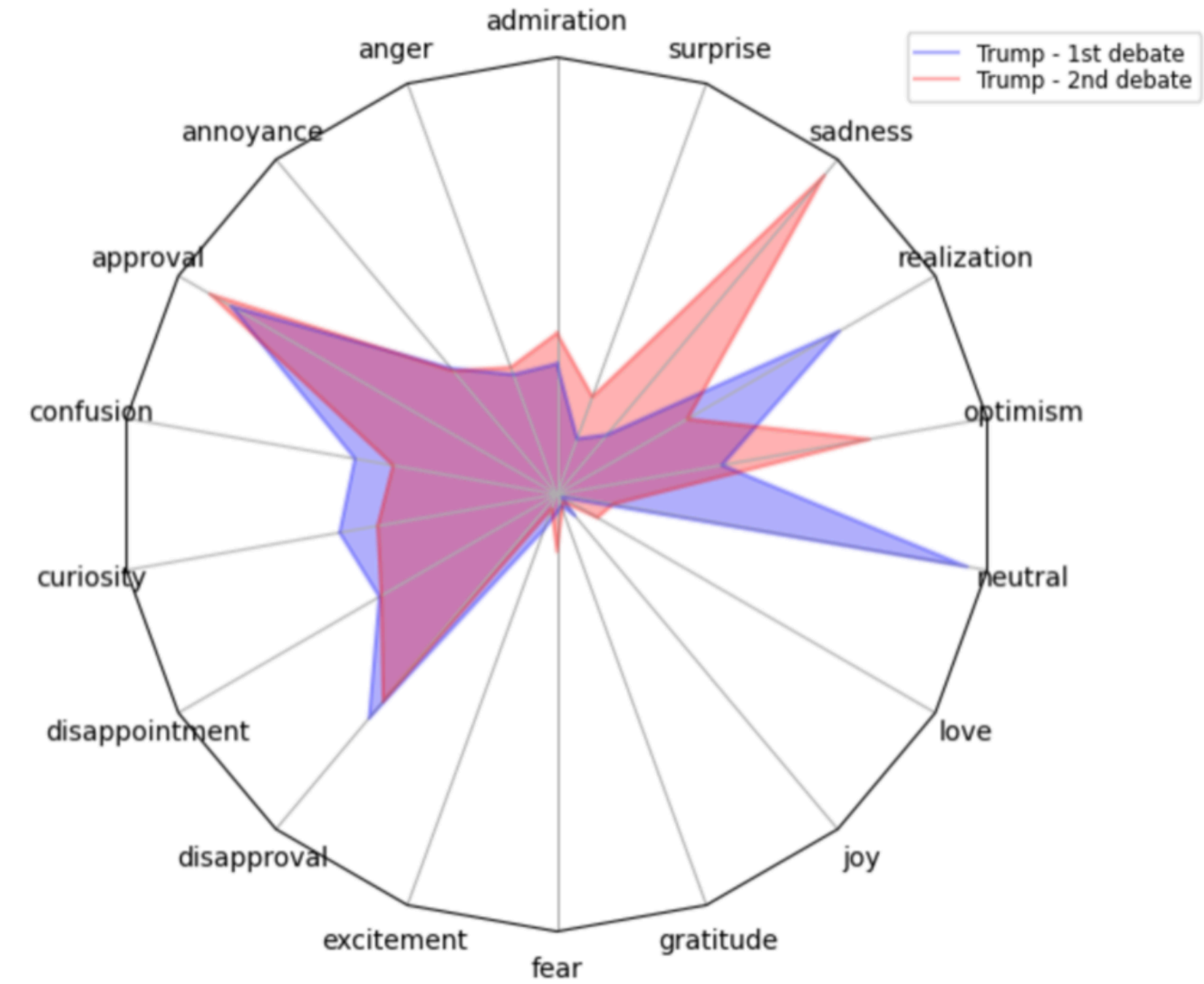
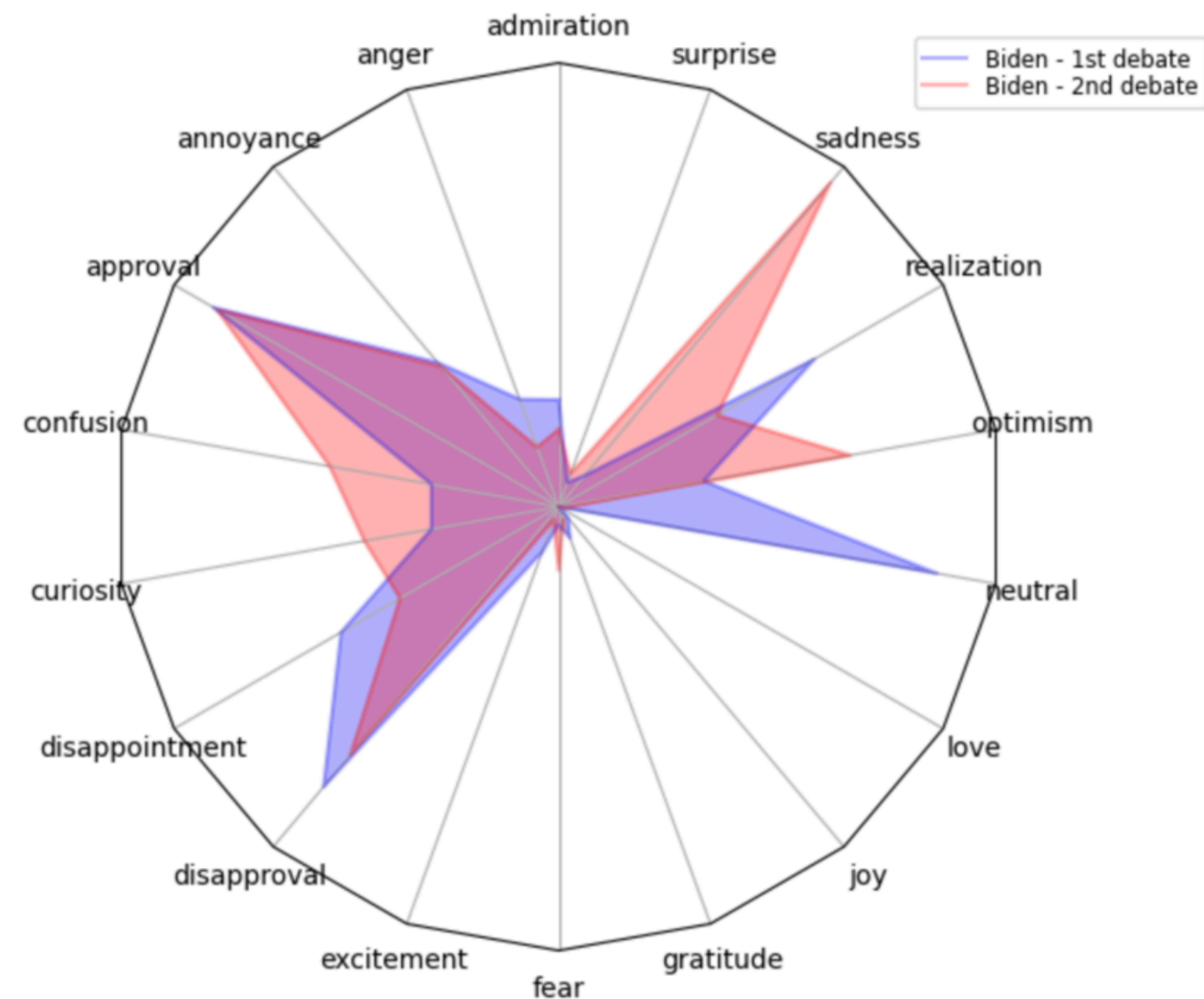


- The emotional language of Donald Trump is more **disperse** and **variable**. Many emotions are short-lived and alternate. Biden's language is more clustered: there are series of emotions.
- Both speakers use **approval** a lot to sound more constructive.
- The usage of **confusion** and **curiosity** is high: Trump uses questions to provoke the opponent.



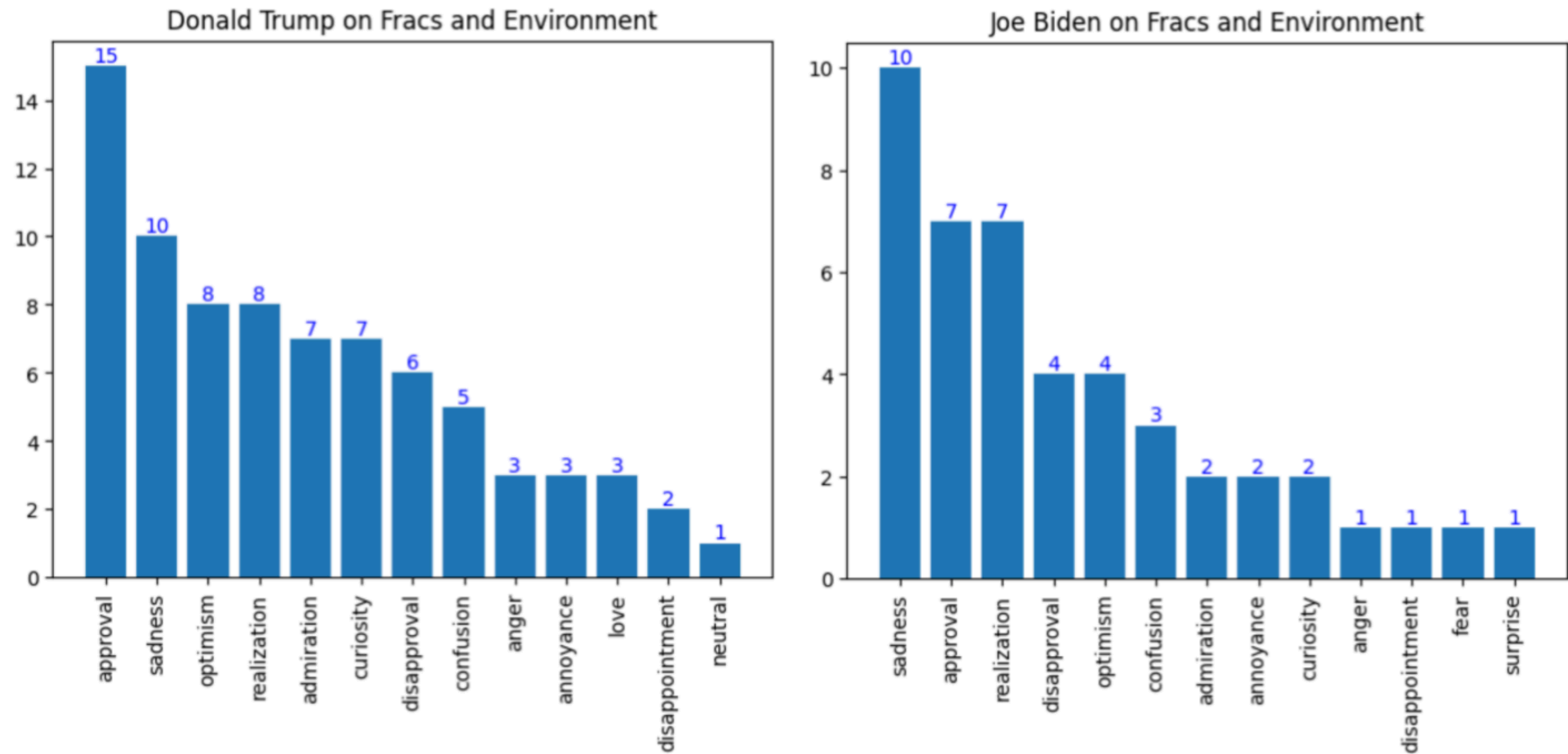
Emotion profile, 2nd presidential debate

Emotional shift comparing the first and second debates



Speakers express **optimism** and **sadness** more. **sadness** - due to the fact that they discussed COVID and it's impact, while **optimism** was used to balance it and form a more **hopeful**, forward-looking message.

Expressed emotions vary across topics



Distribution of expressed emotions during discussion of Fracture and Environment topic

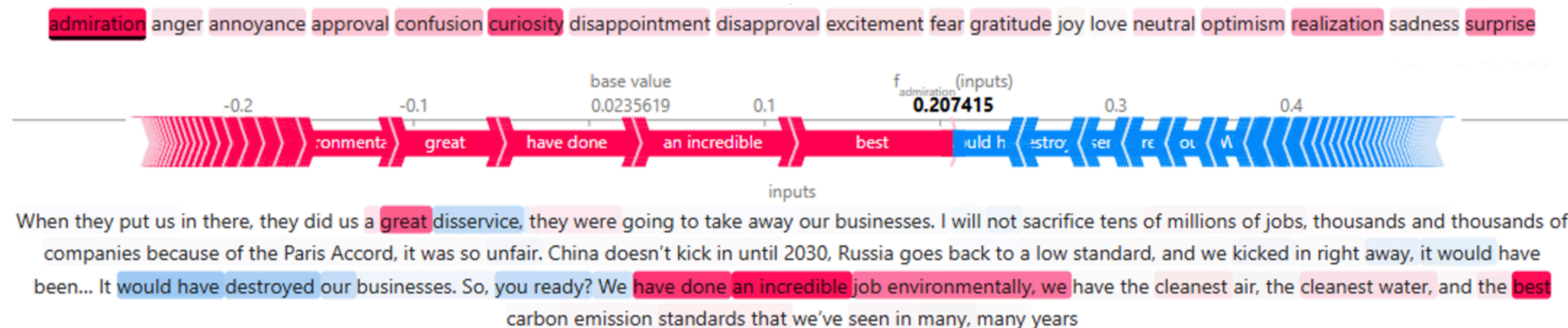
Speakers use different language to persuade the audience

Biden's speech is emotionally charged

- **realization** to evoke personal awakening
- **sadness** and **personal memory** to form emotional appeal and mobilize support.

Trump uses different emotional language

- **approval** to justify his policies and evoke pride
- **admiration** and **optimism** suggests a strategic narrative of success and progress, often appealing to economic wellbeing.



SHAP explainer output

Conclusion

The analysis demonstrated that emotional language is used broadly in political speeches. In the events studied, both Biden and Trump exhibited distinct emotional profiles: Biden's language tended to be more empathetic and thoughtful, while Trump's style appeared more provocative, expressive, and self-assured.

Future work could explore the following directions:

- Improving classifier performance
- Analyzing emotion distribution over longer timeframes (e.g., across multiple years)
- Constructing emotion network graphs or conducting turn-by-turn analyses to examine how one speaker's statements influence the emotional tone of the other
- Investigating co-occurrence patterns among emotions
- Assessing the impact of emotional language on audience responses, such as through comment analysis or post-event surveys